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Movement of molecules from a region of high concentration to a region of low concentration without the expenditure of energy is known as

- ☐ A) Passive diffusion
- ☐ B) Facilitated diffusion
- ☐ C) Active transport
- ☐ D) Group translocation

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Diffusion involving carrier proteins that create channels through the plasma membrane is called

- ☐ A) Passive diffusion
- ☐ B) Facilitated diffusion
- ☐ C) Active transport
- ☐ D) Group translocation

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The rate of transport of molecules across the membrane via facilitated diffusion increases as

- ☐ A) The concentration gradient of the molecule decreases
- ☐ B) The concentration gradient of the molecule increases
- ☐ C) The rate of transport is constant and does not change
- ☐ D) None of the above

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Facilitated diffusion is most prominent in which of the following groups?

- ☐ A) The *Archaea*
- ☐ B) The *Bacteria*
- ☐ C) The *Eukarya*
- ☐ D) All of the above

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_____ is the transport of solute molecules against a concentration gradient with the input of metabolic energy.

- ☐ A) Passive diffusion
- ☐ B) Facilitated diffusion
- ☐ C) Active transport
- ☐ D) Group translocation

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Lactose entering the *E. coli* cell along with a proton is an example of

- ☐ A) Antiport
- ☐ B) Symport
- ☐ C) Passive diffusion
- ☐ D) ABC transporters

25 A population of microbial cells arising from a single cell, representing a single species is called a

- ☐ A) Pure culture
- ☐ B) Mixed culture
- ☐ C) Representative species
- ☐ D) None of the above

26 An appropriate qualitative method for isolating microbes into a pure culture would be

- ☐ A) The streak plate
- ☐ B) The pour plate
- ☐ C) The spread plate
- ☐ D) The serial dilution

27 A microbiology student would like to quantify the number of bacterial cells growing in her sample of nutrient agar. Which of the following methods will help her with this goal?

- ☐ A) Streak plate
- ☐ B) Serial dilution
- ☐ C) Observation of colony morphology
- ☐ D) Metabolic testing

28 The most active bacterial cells on the colony would be found

- ☐ A) Growing along the edge of the colony
- ☐ B) Growing in the center of the colony in contact with the air
- ☐ C) Growing in the center of the colony in contact with the agar
- ☐ D) All cells in the colony will grow equally

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Which of the following is NOT a macronutrient required in large amounts for microbial nutrition?

- ☐ A) Carbon
- ☐ B) Iron
- ☐ C) Phosphorus
- ☐ D) Zinc

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Nutrients required by microbes only in small amounts are called

- ☐ A) Macronutrients
- ☐ B) Micronutrients
- ☐ C) Limited nutrients
- ☐ D) Toxic elements

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An organism that obtains carbon, hydrogen, oxygen, and electrons from organic molecules is a

- ☐ A) Heterotroph
- ☐ B) Autotroph
- ☐ C) Lithotroph
- ☐ D) Chemotroph

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Organisms that use carbon dioxide (CO_2) as their sole or principal source of carbon are called

- ☐ A) Heterotrophs
- ☐ B) Autotrophs
- ☐ C) Lithotrophs
- ☐ D) Chemotrophs

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An organism that uses light for energy and carbon dioxide as a carbon source is a

- ☐ A) Chemoorganoheterotroph
- ☐ B) Chemolithoautotroph
- ☐ C) Photolithoautotroph
- ☐ D) None of the above

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Microbes used industrially to make foods and beverages are

- ☐ A) Chemoorganoheterotrophs
- ☐ B) Chemolithoautotrophs
- ☐ C) Photolithoautotrophs
- ☐ D) None of the above

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Purple nonsulfur and green bacteria are classified as

- ☐ A) Photolithoautotrophs
- ☐ B) Photoorganoheterotrophs
- ☐ C) Chemoorganoheterotrophs
- ☐ D) Chemolithoautotrophs

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Microbes use which of the following processes to obtain nitrogen?

- ☐ A) Metabolism of amino acids
- ☐ B) Incorporation of ammonia directly
- ☐ C) Reduction of nitrate to ammonia
- ☐ D) All of the above

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Phosphorus is required in microbial metabolism for

- ☐ A) Amino acids
- ☐ B) Fatty acids
- ☐ C) Nucleic acids
- ☐ D) Carbohydrates

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Organic compounds that cannot be synthesized by an organism but are essential for growth are called

- ☐ A) Growth factors
- ☐ B) Limited nutrients
- ☐ C) Cofactors
- ☐ D) None of the above

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A major class of growth factors includes

- ☐ A) Amino acids
- ☐ B) Purines and pyrimidines
- ☐ C) Vitamins
- ☐ D) All of the above